DRUGS · HEAD AND NECK ONCOLOGY

Mitomycin C

Mitomycin C is an antitumour antibiotic that, applied topically for a few minutes, inhibits fibroblast proliferation. ENT surgeons apply it after endoscopic treatment of laryngotracheal stenosis, choanal atresia repair, dacryocystorhinostomy and frontal sinus surgery to reduce restenosis. The evidence is mixed, and its effect depends on technique.

1 CHAPTER ONE At a glance

The drug on one page.

1.1 Summary

MITOMYCIN C

	SUMMARY
Class	Alkylating antitumour antibiotic
How it works	Cross-links DNA and inhibits fibroblast proliferation and collagen formation
ENT uses	Adjunct after endoscopic incision and dilatation of subglottic and tracheal stenosis; choanal atresia repair; endoscopic dacryocystorhinostomy; frontal sinus and other stenoses; glottic webs
Dose	Topical 0.4 to 0.5 mg/ml on a cottonoid or pledget for 2 to 5 minutes, then irrigation with saline
Handling	Cytotoxic: prepare and dispose of according to cytotoxic protocols; avoid contact with skin and eyes

2 CHAPTER TWO Using it in ENT

Practice.

2.1 Airway stenosis

After radial incisions and balloon dilatation of an idiopathic or post-intubation subglottic stenosis, topical mitomycin may lengthen the interval between procedures in some series; randomised evidence is limited. Inhaled steroids, co-trimoxazole in some idiopathic cases, and treatment of reflux are other adjuncts.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

Excessive fibrinous debris or granulation that can obstruct a small airway (check endoscopically), delayed healing, and theoretical carcinogenicity; systemic absorption is negligible at topical doses.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Mechanism and concentration.
- Where it is used and the limits of the evidence.

4.2 Common pitfalls

COMMON PITFALLS

- Leaving fibrinous debris in a child's airway after mitomycin.

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